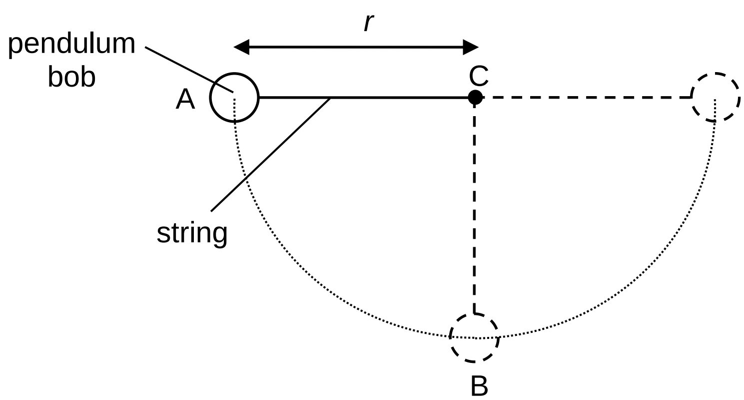
9	A pendulum bob of mass 0.100 kg is supported by a string and swung along a circular path of radius *r* about the fixed point C. The bob is momentarily at rest at point A, with the string horizontal and just taut.



What is the tension in the string when the bob is at point B which is vertically below point C?

A	0.981 N	**B**	1.96 N	**C**	2.94 N	**D**	3.92 N

Answer: C

Given that at point A the string is horizontal & just taut → T = 0 at point A

Since Weight acts vertically, it cannot provide for the centripetal force at point A.

So at point A, centripetal force = 0 → speed at point A must be zero.

By principle of conservation of energy,
Gain in KE from A to B = Loss in GPE from A to B

$$\left(\frac{1}{2}mv^2\right)_{\text{point B}} - 0 = mgr$$

At point B, $v^2 = 2gr$ ---- (1)

At point B, the resultant force will be towards the centre of the vertical circle is

$$T - mg = \frac{mv^2}{r}$$

$$T = \frac{mv^2}{r} + mg \quad \text{----(2)}$$

Sub (1) into (2):

$$T = \frac{m(2rg)}{r} + mg = 3mg = 3(0.100)(9.81) = 2.943 = 2.94\ \text{N}$$

10	At a point on the surface of a uniform sphere of diameter d, the gravitational field strength due